

**AMENDMENT NO. 2 SEPTEMBER 2000**  
**TO**  
**IS 9300 ( PART 2 ) : 1989 STATISTICAL MODELS FOR**  
**INDUSTRIAL APPLICATIONS**  
**PART 2 CONTINUOUS MODELS**  
**( First Revision )**

( Page 7, clause 7.4 ) — Insert the following clause at the end of 7.4:

**7.5 Fitting a Gamma Model**

**7.5.1 Example —** In a manufacturing process of jute proudcts, breaker card stage is the first stage of filamentation for the subsequent processing. From a sample of 10 cm carded sliver ( strand of carded raw jute is called sliver), single fibres were segregated and their lengths were measured which are grouped in a frequency table ( see Table 6 ).

Fit a gamma model to the above data and test its goodness of fit.

**Table 6 Length of Fibres (mm) of Carded Silver**

| Class Interval | Frequency    |
|----------------|--------------|
| 0 -5           | 350          |
| 5 -10          | 575          |
| 10 -15         | 500          |
| 15 -20         | 325          |
| 20 -25         | 215          |
| 25 -30         | 135          |
| 30 -35         | 50           |
| 35-40          | 25           |
| 45-50          | 5            |
| <u>Total</u>   | <u>2 190</u> |

**Amend No. 2 to IS 9300 (Part 2) : 1989**

The mean  $\bar{x}$  and variance  $s^2$  as calculated from the frequency table are:

$$\bar{x} = 13.20 \text{ mm and } s^2 = 72.63 \text{ mm}$$

The parameters of the gamma model  $\eta$  and  $\lambda$  can be calculated by solving the following conditions:

$$\begin{aligned} \bar{x} &= \eta/\lambda \text{ and } s^2 = \eta/\lambda^2 \\ \lambda &= \bar{x}/s^2 = 13.20/72.63 = 0.1817 \\ \text{and } \eta &= \lambda \bar{x} = 2.399 = 2.40 \text{ (approx)} \end{aligned}$$

The next step is to calculate the expected frequencies ( $e_i$ ) based on the assumption that the above frequency distribution is coming from a gamma distribution. The steps are described in Table 7.

The probability for each class interval is obtained from the table of the Incomplete Gamma Function.

**Table 7 Expected Frequencies Based on Gamma Distribution**

| Class Interval<br>( $x_L - x_U$ ) | $x$ | $\Gamma(0.1173x, 1.4)$ | Probability<br>( $x_L < x < x_U$ ) | Expected<br>Frequency( $e_i$ ) |
|-----------------------------------|-----|------------------------|------------------------------------|--------------------------------|
| (1)                               | (2) | (3)                    | (4)                                | (5) = 2190x(4)                 |
| 0-5                               | 5   | 0.149                  | 0.149                              | 326.31                         |
| 5-10                              | 10  | 0.424                  | 0.275                              | 602.25                         |
| 10-15                             | 15  | 0.662                  | 0.238                              | 521.22                         |
| 15-20                             | 20  | 0.815                  | 0.153                              | 335.07                         |
| 20-25                             | 25  | 0.901                  | 0.086                              | 188.34                         |
| 25-30                             | 30  | 0.955                  | 0.054                              | 118.26                         |
| 30-35                             | 35  | 0.977                  | 0.022                              | 48.18                          |
| 35-40                             | 40  | 0.989                  | 0.012                              | 26.28                          |
| 40-45                             | 45  | 0.995                  | 0.006                              | 13.14                          |
| 45-50                             | 50  | 0.998                  | 0.003                              | 6.57                           |

**Amend No. 2 to IS 9300 (Part 2) : 1989**

**Goodness of Fit from  $\chi^2$  Test :** After calculating expected frequencies ( $e_i$ ) for each class interval, their closeness with the observed frequencies ( $o_i$ ) are tested with the help of  $\chi^2$  test (see Table 8).

| Table 8 Calculations for This $\chi^2$ Test |                                |                              |                       |
|---------------------------------------------|--------------------------------|------------------------------|-----------------------|
| Class Interval                              | Observed Frequencies ( $O_i$ ) | Expected Frequency ( $e_i$ ) | $(O_i - e_i)^2 / e_i$ |
| 0-5                                         | 350                            | 326.31                       | 1.72                  |
| 5-10                                        | 575                            | 602.22                       | 1.23                  |
| 10-15                                       | 500                            | 521.22                       | 0.86                  |
| 15-20                                       | 325                            | 335.07                       | 0.30                  |
| 20-25                                       | 215                            | 188.34                       | 3.77                  |
| 25-30                                       | 135                            | 118.26                       | 2.37                  |
| 30-35                                       | 50                             | 48.18                        | 0.07                  |
| 35-40                                       | 25                             | 26.28                        | 0.06                  |
| 40-45                                       | 10                             | 13.14                        | 0.75                  |
| 45-50                                       | 5                              | 6.57                         | 0.37                  |
| Total $\chi^2 = 11.50$                      |                                |                              |                       |

The total number of classes is 10. Three degrees of freedom are apportioned for the estimation of mean, standard deviation and for total frequency. Thus calculated value of  $\chi^2$  is compared with the tabulated value [see IS 6200 (Part 2) : 1977 ] of  $\chi^2 = 14.07$  for 7 degrees of freedom at 5 percent level of significance. Since the calculated value is less than the tabulated value the fit can be taken as good one.

( Page 8, Tables 6, 7, 8 and 9 ) — Table 6, Table 7, Table 8, Table 9 may be replaced by Table 9, Table 10, Table 11, Table 12 respectively.

**Amend No. 2 to IS 9300 ( Part 2 ) : 1989**

**( Page 8, clause 8.5 )** — In line 5, reference to Table 6 may be replaced by Table 9.

**( Page 8, clause 8.5.1 )** — In line 3, reference to Table 7 may be replaced by Table 10.

**( Page 8, clause 8.5.2 )** — In line 4, reference to Table 8 may be replaced by Table 11.

**( Page 8, clause 8.5.3 )** — In line 2, reference to Table 9 may be replaced by Table 12.

**( Page 1, clause 9.4.1, Amendment No. 1 )** — In line 15, reference to Table 10 may be replaced by Table 13.

**( Page 2, clause 9.4.1, Amendment No. 1 )** — Reference to Table 11 may be replaced by Table 14.

**( Page 2, Table 10, Amendment No. 1 )** — Table 10 may be replaced by Table 13.

**( Page 3, Table 11, Amendment No. 1 )** — Table 11 may be replaced by Table 14.

**( MSD 3 )**